

DBMS – ORDER AND ORDER_ITEM MANAGEMENT

AJIO E-Commerce database management system

1.DATABASE DESIGN

1.1 ORDER TABLE

Attribute	Description
Order_ID	Unique order ID
Customer_ID	Customer ID
Order_Date	Date of order
Total_Amount	Total order amount
Order_Status	Current order status
Shipping_Address	Delivery address
Delivery_Date	Expected delivery date

1.2 ORDER_ITEM TABLE

Attribute	Description
Order_Detail_ID	Unique order item ID
Order_ID	Order ID
Product_ID	Product ID
Quantity	Quantity ordered
Unit_Price	Price per product
Total_Price	Total price

2.Relationship Between Order and Order_Item Tables

- **Orders.Order_ID** – Primary Key in the Orders table.
- **Order_Items.Order_Item_ID** – Primary Key in the Order Items table.
- **Order_Items.Order_ID** – Foreign Key referencing `Orders (Order_ID)`.
- **Orders → Order_Items** – One-to-Many (1:M) relationship.

3.TABLE CREATION

3.1 ORDER TABLE

```

SQL> CREATE TABLE Orders (
2     Order_ID NUMBER PRIMARY KEY,
3     Customer_ID NUMBER,
4     Order_Date DATE,
5     Total_Amount NUMBER(10,2),
6     Order_Status VARCHAR2(30),
7     Shipping_Address VARCHAR2(200),
8     Delivery_Date DATE
9 );

```

Table created.

3.2 ORDER_ITEM TABLE

```

SQL> CREATE TABLE Order_Items (
2     Order_Item_ID NUMBER PRIMARY KEY,
3     Order_ID NUMBER,
4     Product_ID NUMBER,
5     Quantity NUMBER,
6     Unit_Price NUMBER(10,2),
7     Total_Price NUMBER(10,2),
8     FOREIGN KEY (Order_ID) REFERENCES Orders(Order_ID)
9 );

```

Table created.

4. INSERTION

4.1 Inserting values into Order table

```

SQL> INSERT INTO Orders VALUES
2 (101, 1, TO_DATE('10-09-2026','DD-MM-YYYY'), 2499.00, 'Delivered', 'Chennai, Tamil Nadu', TO_DATE('14-09-2026','DD-MM-YYYY'));
1 row created.

SQL>
SQL> INSERT INTO Orders VALUES
2 (102, 2, TO_DATE('11-09-2026','DD-MM-YYYY'), 1799.00, 'Shipped', 'Bangalore, Karnataka', TO_DATE('15-09-2026','DD-MM-YYYY'));
1 row created.

SQL>
SQL> INSERT INTO Orders VALUES
2 (103, 3, TO_DATE('12-09-2026','DD-MM-YYYY'), 3299.00, 'Processing', 'Coimbatore, Tamil Nadu', TO_DATE('17-09-2026','DD-MM-YYYY'));
1 row created.

SQL>
SQL> INSERT INTO Orders VALUES
2 (104, 4, TO_DATE('13-09-2026','DD-MM-YYYY'), 1499.00, 'Delivered', 'Madurai, Tamil Nadu', TO_DATE('17-09-2026','DD-MM-YYYY'));
1 row created.

SQL>
SQL> INSERT INTO Orders VALUES
2 (105, 5, TO_DATE('14-09-2026','DD-MM-YYYY'), 2899.00, 'Shipped', 'Hyderabad, Telangana', TO_DATE('18-09-2026','DD-MM-YYYY'));
1 row created.

```

4.1 Inserting values into Order_item table

```
SQL> INSERT INTO Order_Items VALUES
  2  (1, 101, 201, 2, 799.00, 1598.00);

1 row created.

SQL>
SQL> INSERT INTO Order_Items VALUES
  2  (2, 101, 202, 1, 901.00, 901.00);

1 row created.

SQL>
SQL> INSERT INTO Order_Items VALUES
  2  (3, 102, 203, 1, 1799.00, 1799.00);

1 row created.

SQL>
SQL> INSERT INTO Order_Items VALUES
  2  (4, 103, 204, 2, 1299.50, 2599.00);

1 row created.

SQL>
SQL> INSERT INTO Order_Items VALUES
  2  (5, 103, 205, 1, 700.00, 700.00);

1 row created.

SQL>
SQL> INSERT INTO Order_Items VALUES
  2  (6, 104, 206, 1, 1499.00, 1499.00);

1 row created.

SQL>
SQL> INSERT INTO Order_Items VALUES
  2  (7, 105, 207, 2, 999.50, 1999.00);
```

5. MODIFICATION

5.1 Before Modifying Order_Status

```
SQL> SELECT * FROM Orders
2  WHERE Order_ID = 103;

ORDER_ID CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
-----
SHIPPING_ADDRESS
-----
DELIVERY_
-----
      103           3 12-SEP-26          3299 Processing
Coimbatore, Tamil Nadu
17-SEP-26
```

5.2 Modifying Order_Status

```
SQL> UPDATE Orders
2  SET Order_Status = 'Shipped'
3  WHERE Order_ID = 103;

1 row updated.
```

5.3 After Modifying Order_Status

```
SQL> SELECT * FROM Orders
2  WHERE Order_ID = 103;

ORDER_ID CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
-----
SHIPPING_ADDRESS
-----
DELIVERY_
-----
      103           3 12-SEP-26          3299 Shipped
Coimbatore, Tamil Nadu
17-SEP-26
```

6. MODIFYING ORDER_ITEMS TABLE

6.1 Before Modifying Order_Items table

```
SQL> SELECT * FROM Order_Items
2  WHERE Order_Item_ID = 1;
```

ORDER_ITEM_ID	ORDER_ID	PRODUCT_ID	QUANTITY	UNIT_PRICE	TOTAL_PRICE
1	101	201	2	799	1598

6.2 Modifying Order_Items table

```
SQL> UPDATE Order_Items
2  SET Quantity = 3,
3      Total_Price = 2397.00
4  WHERE Order_Item_ID = 1;
```

```
1 row updated.
```

6.3 After Modifying Order_Items table

```
SQL>
SQL> SELECT * FROM Order_Items
2  WHERE Order_Item_ID = 1;
```

ORDER_ITEM_ID	ORDER_ID	PRODUCT_ID	QUANTITY	UNIT_PRICE	TOTAL_PRICE
1	101	201	3	799	2397

7. Order_item table After modification

```
SQL> SELECT * FROM Order_Items;
```

ORDER_ITEM_ID	ORDER_ID	PRODUCT_ID	QUANTITY	UNIT_PRICE	TOTAL_PRICE
1	101	201	2	799	1598
2	101	202	1	950	950
3	102	203	1	1799	1799
4	103	204	2	1299.5	2599
5	103	205	1	700	700
6	104	206	1	1499	1499
7	105	207	2	999.5	1999
8	105	208	1	900	900

```
8 rows selected.
```

8. Customer Order History Report

CUSTOMER_ID	ORDER_ID	ORDER_DAT	PRODUCT_ID	QUANTITY	UNIT_PRICE	TOTAL_PRICE
1	101	10-SEP-26	201	2	799	1598
2499	Delivered					

1	101	10-SEP-26	202	1	950	950
2499	Delivered					
2	102	11-SEP-26	203	1	1799	1799
1799	Delivered					

CUSTOMER_ID	ORDER_ID	ORDER_DAT	PRODUCT_ID	QUANTITY	UNIT_PRICE	TOTAL_PRICE
3	103	12-SEP-26	204	2	1299.5	2599
3299	Processing					

3	103	12-SEP-26	205	1	700	700
3299	Processing					
4	104	13-SEP-26	206	1	1499	1499
1499	Delivered					

CUSTOMER_ID	ORDER_ID	ORDER_DAT	PRODUCT_ID	QUANTITY	UNIT_PRICE	TOTAL_PRICE
5	105	14-SEP-26	207	2	999.5	1999
2899	Shipped					
5	105	14-SEP-26	208	1	900	900
2899	Shipped					